BILL NO. 2 FACTORY AND ADMINISTRATION OFFICE BUILDING

ITEM NO.	ITEM DESCRIPTION	Unit	Qty	Unit Rate	Total Amount (QR)	
10	**ACMV WORK**					
	AIR CONDITIONING WORK					
	Supply, installation, testing, and commissioning of VRF outdoor condensing units complete with refrigerant copper piping network, Y-Branch fittings and all other fittings, thermal insulation for all refrigerant pipes using high-quality elastomeric closed-cell insulation wrapped with protective fiber cloth and finished with a UV-resistant protective coating, fabricated powder-coated steel supports with anti-vibration pads, proper alignment and leveling, interconnection with indoor units, Concrete foundation, power cabling and control wiring, refrigerant leak testing, evacuation, charging, and final commissioning, all executed in full compliance with ASHRAE standards, manufacturer's recommendations, and project specifications.					
	BRAND : DAIKIN INDIA					
A	VRF set-01, 177.80 Kw	nos.	1	202285	QAR	202,285.00
B	VRF set-02, 168.60 Kw	nos.	1	192685	QAR	192,685.00
C	VRF set-03, 231.00 Kw	nos.	1	264000	QAR	264,000.00
D	VRF set-04, 276.10 Kw	nos.	1	315542	QAR	315,542.00
					QAR	-
	Supply, installation, testing, and commissioning of indoor units for VRF system including FCU and wall mounted units complete with motors, cooling coils, filters (where applicable), drain pans, control valves, refrigerant and drain piping to the nearest drain point, thermal insulation, supports, and all necessary accessories, in full compliance with ASHRAE standards, manufacturer's recommendations, and project specifications. The work includes proper alignment and leveling of all units, fabrication and installation of hangers, brackets, and vibration isolators, interconnection with refrigerant piping network and control wiring, drain lines with proper slope and insulation to prevent condensation, testing for performance and leakage, final commissioning witnessed by the consultant, and bilingual labeling (Arabic and English) for identification and maintenance, ensuring that the units meet the required capacity, static pressure, efficiency.				QAR	-
					QAR	-
E	FCU-01 - 2.8Kw	nos.	6	1000	QAR	6,000.00
F	FCU-02 - 3.6Kw	nos.	2	1028	QAR	2,056.00
G	FCU-03 - 4.3Kw	nos.	3	1228	QAR	3,684.00
H	FCU-04 - 5.0Kw	nos.	3	1428	QAR	4,284.00
I	FCU-05 - 5.6Kw	nos.	4	1600	QAR	6,400.00
J	FCU-06 - 6.3Kw	nos.	4	1800	QAR	7,200.00
K	FCU-07 - 7.1Kw	nos.	6	2028	QAR	12,168.00
L	FCU-08 - 8.4Kw	nos.	3	2400	QAR	7,200.00
					QAR	-
					QAR	-
		Carried to Collection QR			**QAR**	**1,023,504.00**

ITEM NO.	ITEM DESCRIPTION	Unit	Qty	Unit Rate	Total Amount (QR)	
10	**ACMV WORK, CONTINUES.**					
A	FCU-09 - 11.2Kw	nos.	14	3200	QAR	44,800.00
B	FCU-10 - 14.2Kw	nos.	10	4057	QAR	40,570.00
C	FCU-11 - 16.0Kw	nos.	6	4571	QAR	27,426.00
D	FCU-12 - 9.0Kw	nos.	2	2571	QAR	5,142.00
E	FCU-13 - 22.4Kw	nos.	11	6400	QAR	70,400.00
F	FCU-14 - 2.2Kw	nos.	2	1000	QAR	2,000.00
G	WMU-01 - 2.2Kw	nos.	1	1000	QAR	1,000.00
H	WMU-02 - 4.3Kw	nos.	1	1228	QAR	1,228.00
I	WMU-03 - 7.1Kw	nos.	1	2028	QAR	2,028.00
J	WMU-04 - 9.0Kw	nos.	1	2571	QAR	2,571.00
					QAR	-
	Supply, installation, testing, and commissioning of DX Fresh Air Handling Unit (FAHU), complete with double-skin thermally insulated casing, DX cooling coil and refrigerant circuit, compressor section, fan and motor assembly, stainless steel drain pan, fresh air intake section, pre-filters and fine filters, access doors, integrated control panel, vibration isolators, sound attenuator, dampers, and all required accessories, including proper positioning on the roof or designated mechanical area, connection to refrigerant piping and ductwork, installation of supports and vibration control, provision of electrical power and control wiring, integration with BMS where applicable, testing for refrigerant pressure and leak detection, airflow testing and balancing, and full functional commissioning, all in full compliance with ASHRAE standards, SMACNA guidelines, manufacturer's recommendations, project specifications, and the approved shop drawings.				QAR	-
	BRAND : DAIKIN INDIA				QAR	-
K	FAHU-01, 2800CFM	nos.	1	66900	QAR	66,900.00
					QAR	-
	Supply, installation, testing, and commissioning of exhaust fans, complete with louver and all necessary accessories, including proper wall opening, mounting, connection to power supply, and final testing to ensure proper operation, all in accordance with ASHRAE standards, manufacturer's recommendations, and project specifications.				QAR	-
	BRAND : MCKENNLY QATAR				QAR	-
L	Inline ducted, Exhaust air fan (75CFM) EXF-01	nos.	1	500	QAR	500.00
M	Inline ducted, Exhaust air fan (300CFM) EXF-02	nos.	4	2,000	QAR	8,000.00
N	Inline ducted, Exhaust air fan (225CFM) EXF-03	nos.	1	2,000	QAR	2,000.00
O	Inline ducted, Exhaust air fan (275CFM) EXF-04	nos.	1	2,000	QAR	2,000.00
P	Inline ducted, Exhaust air fan (450CFM) EXF-05	nos.	1	3,500	QAR	3,500.00
Q	Inline ducted, Exhaust air fan (150CFM) EXF-06	nos.	1	2,000	QAR	2,000.00
R	Wall mounted, Exhaust air fan (75CFM) EXF-07	nos.	4	250	QAR	1,000.00
S	Wall mounted, Exhaust air fan (150CFM) EXF-08	nos.	2	250	QAR	500.00
T	Inline ducted, Exhaust air fan (525CFM) EXF-09	nos.	1	3,500	QAR	3,500.00
					QAR	-
					QAR	-
		Carried to Collection QR			**QAR**	**287,065.00**

ITEM NO.	ITEM DESCRIPTION	Unit	Qty	Unit Rate	Total Amount (QR)
10	**ACMV WORK, CONTINUES.**				
A	For grilles installed in the Administration Office Building, the contractor shall carry out full coordination with the interior design team and the client to confirm grille locations, types, sizes, orientations, and final appearance prior to installation. Supply, installation, testing, and commissioning of air grilles (supply, return, and exhaust) complete with volume control dampers (VCD) where required, flexible ducts, frames, mounting accessories, and proper alignment, including cutting of openings, secure fixing to ducts or ceiling/wall surfaces, connection to the respective duct system, adjustment for proper airflow, and final air balancing, all in accordance with ASHRAE standards, SMACNA guidelines, manufacturer's recommendations, project specifications, and as shown on the approved shop drawings.	item			
B	150x150, Exhaust air grill	nos.	31	100	QAR 3,100.00
C	300x300, Return air grill	nos.	122	100	QAR 12,200.00
D	300x300, Supply air grill	nos.	122	100	QAR 12,200.00
E	300x300, Side wall return air grill	nos.	4	100	QAR 400.00
F	300x300, Side wall supply air grill	nos.	4	100	QAR 400.00
G	25slot width 3slots, Supply linear bar grill	m	23	300	QAR 6,900.00
H	25slot width 3slots, Retuen linear bar grill	m	22	250	QAR 5,500.00
		Carried to Collection QR			**QAR 40,700.00**

ITEM NO.	ITEM DESCRIPTION	Unit	Qty	Unit Rate	Total Amount (QR)	
10	**ACMV WORK, CONTINUES.**					
	Supply, installation, testing, and commissioning of exhaust air ducts fabricated from galvanized steel sheets, complete with all required fittings, flanges, gaskets, sealants, and accessories, including proper fabrication in accordance with SMACNA standards, installation with hangers and supports, sealing of all joints to prevent leakage, connection to exhaust fans and grilles, and performing final testing and air balancing to ensure the specified airflow, all in full compliance with ASHRAE standards, SMACNA guidelines, project specifications, and as shown on the approved shop drawings.					
. A	Exhaust air duct	m^2	193	120	QAR	23,160.00
					QAR	-
	Supply, installation, testing, and commissioning of fresh air and supply air ducts fabricated from galvanized steel sheets, complete with all required fittings, flanges, gaskets, sealants, insulation materials, and accessories, including fabrication as per SMACNA standards, installation with hangers and supports, sealing of all joints, connection to fans, AHU/FAHU units, and grilles, and performing final testing and air balancing to ensure the specified airflow; all ducts shall be fully insulated with approved thermal insulation (fiberglass or elastomeric) of the specified thickness with continuous vapor barrier and protective finish where required, all in full compliance with ASHRAE standards, SMACNA guidelines, project specifications, and as shown on the approved shop drawings.				QAR	-
					QAR	-
B	Fresh air duct	m^2	256	200	QAR	51,200.00
C	Supply air duct	m^2	585	200	QAR	117,000.00
					QAR	-
	Supply, installation, testing, and commissioning of dampers including Volume Control Dampers (VCD), Fire Dampers (FD), and Smoke Dampers (SD), complete with frames, blades, actuators (where required), fusible links, access doors, mounting accessories, sealing materials, and proper alignment, including cutting of openings, secure fixing to ductwork or wall/partition systems, providing access for maintenance, connection to control systems when applicable, and performing final testing and balancing; all works shall be carried out in full compliance with NFPA standards, UL/FM requirements, ASHRAE standards, SMACNA guidelines, project specifications, and as shown on the approved shop drawings.				QAR	-
					QAR	-
D	VCD 400X400	nos.	1	250	QAR	250.00
E	VCD 250X200	nos.	7	150	QAR	1,050.00
F	VCD 300X200	nos.	8	150	QAR	1,200.00
G	VCD 100X100	nos.	3	150	QAR	450.00
		Carried to Collection QR			**QAR**	**194,310.00**

ITEM NO.	ITEM DESCRIPTION	Unit	Qty	Unit Rate	Total Amount (QR)	
10	**ACMV WORK, CONTINUES.**					
A	VCD 100X150	nos.	12	150	QAR	1,800.00
B	VCD 250X250	nos.	1	150	QAR	150.00
C	VCD 250X300	nos.	1	150	QAR	150.00
D	VCD 150X200	nos.	1	150	QAR	150.00
E	VCD 150X175	nos.	1	150	QAR	150.00
F	VCD 150X150	nos.	25	150	QAR	3,750.00
G	VCD 200X100	nos.	2	150	QAR	300.00
H	VCD 200X150	nos.	3	150	QAR	450.00
I	VCD 300X350	nos.	1	150	QAR	150.00
J	VCD 500X450	nos.	1	150	QAR	150.00
K	SD 500X450	nos.	1	150	QAR	150.00
L	SD 400X400	nos.	1	150	QAR	150.00
M	SD 300X200	nos.	2	150	QAR	300.00
O	FD 250X200	nos.	1	1,100	QAR	1,100.00
P	FD 500X500	nos.	1	1,100	QAR	1,100.00
Q	FD 150X150	nos.	4	1,100	QAR	4,400.00
R	FD 300X200	nos.	2	1,100	QAR	2,200.00
S	FD 200X150	nos.	1	1,100	QAR	1,100.00
					QAR	-
	Supply, installation, testing, and commissioning of **new dehumidifiers**, complete with internal compressor unit, coil assembly, fan section, drain connection, humidistat/temperature control interface where applicable, filters, electrical power wiring, supports, mounting accessories, and all required fittings, including proper placement as per room humidity requirements, connection to the drainage system, and full functional testing to verify moisture removal capacity and automatic control operation, all in full compliance with manufacturer's recommendations, ASHRAE standards, project specifications, and the approved shop drawings.				QAR	-
					QAR	-
T	New protable dehumidifier, with a capacity of 35L/d@ 18℃	nos.	5	9500	QAR	47,500.00
		Carried to Collection QR			**QAR**	**65,200.00**

ITEM NO.	ITEM DESCRIPTION	Unit	Qty	Unit Rate	Total Amount (QR)
10	**ACMV WORK, CONTINUES.** **VENTELATION WORK**				
	Supply, installation, testing, and commissioning of smoke duct detector, complete with sampling tubes, control module, mounting accessories, brackets, wiring connections, conduit provisions, and all required fittings, including proper installation on the designated duct sections, sealing around penetrations, alignment to airflow direction, functional testing, integration with the fire alarm control panel (FACP), and verification of alarm/trip signals; all works shall be in full compliance with NFPA 72, NFPA 90A, QCDD requirements, manufacturer's recommendations, project specifications, and as shown on the approved shop drawings.				
A	Smoke duct detector	nos.	1	EXCL	
	Supply, installation, testing, and commissioning of electrically actuated roof smoke/heat vents, complete with insulated hatch panels, curb assembly, electric actuator mechanism, gaskets, hinges, supports, mounting accessories, and all hardware, including proper installation on roof upstands with airtight and watertight sealing, full electrical integration with the Fire Alarm Control Panel (FACP) for automatic opening upon fire alarm activation, connection to the emergency power supply to ensure operation during power failure, installation of all wiring and control circuits using QCDD-approved fire-resistant cables, and complete functional testing of actuation and feedback signals, all in full compliance with NFPA 204, QCDD requirements, manufacturer's recommendations, project specifications, and the approved shop drawings.				
B	(0.8mx3m) Smoke/heat vents	nos.	24	EXCL	
		Carried to Collection QR			

ITEM NO.	ITEM DESCRIPTION	Unit	Qty	Unit Rate	Total Amount (QR)
10	**ACMV WORK, CONTINUES.**				
	Supply, installation, testing, and commissioning of ventilation louvers complete with motorized dampers, including aluminum/galvanized louver assemblies, damper blades, electric actuator, bird screen where required, frames, sealing materials, supports, and mounting accessories, with proper installation on wall or shaft openings ensuring accurate alignment and weatherproof sealing, electrical interfacing of the damper actuator with the Fire Alarm Control Panel (FACP) for automatic operation in fire/emergency modes, connection to the emergency power supply for uninterrupted functionality, installation of all control and power cabling using QCDD-approved fire-resistant cables, and complete functional testing and verification, all in full compliance with NFPA 90A, NFPA 70, QCDD requirements, ASHRAE standards, SMACNA guidelines, manufacturer's recommendations, project specifications, and the approved shop drawings.				
A	2.5X3.0 LOUVER WITH MOTORIZED DAMPER	nos.	8	8000	QAR 64,000.00
B	2.5X2.0 LOUVER WITH MOTORIZED DAMPER	nos.	8	6500	QAR 52,000.00
	excluding power by electrical contractor				QAR -
	Supply, installation, testing, and commissioning of MCC panel and Override panel complete with all required components, wiring, controls, power supply, labeling, and accessories, including proper mounting, termination, programming, and integration with smoke control fans, motorized dampers, and other life safety equipment, with manual override functions provided for emergency operation, and final testing, simulation, and commissioning witnessed by the consultant and relevant authorities, all in full compliance with NFPA 92, QCDD regulations, manufacturer's recommendations, and project specifications.	Item	1	25000	QAR 25,000.00
					QAR -
	SUNDRIES (COMPRESSED AIR)				QAR -
					QAR -
C	Tagging & Labelling	Item	1	5000	QAR 5,000.00
D	O & M, As- built drawings	Item	1	5000	QAR 5,000.00
E	Testing & Commissioning	Item	1	25000	QAR 25,000.00
	ADJUSTMENT ITEMS				
	The contractor to provide the adjustment items / missing item required to complete the work according to the consultant requirement / Athurity requirement.				
	Protection				
F	Allow for protecting all the work in this section	lot	1		
G	Allow for any other necessary items required for the completion of the work as described in the relevant sections of the drawings or specifications and not mentioned herein 1 item	lot	1		
		Carried to Collection QR			**QAR 176,000.00**

ITEM NO.	ITEM DESCRIPTION	Unit	Qty	Unit Rate	Total Amount (QR)
10	ACMV WORK, CONTINUES. **BILL NO. 2 FACTORY AND ADMINISTRATION OFFICE BUILDING COLLECTION** Total from Page 2/10/1 Total from Page 2/10/2 Total from Page 2/10/3 Total from Page 2/10/4 Total from Page 2/10/5 Total from Page 2/10/6 Total from Page 2/10/7 except smoke fans/smoke control				
	Total For AC & Ventilation	**To Bill No 2 Summary**			**QAR 1,786,779.00**